



Algebra 2 Curriculum Map

SY 2025-2026

ALIGNED TO:

2017 MASSACHUSETTS CURRICULUM FRAMEWORK FOR MATHEMATICS

SAVVAS enVISION GEOMETRY

[\(CTRL+Click to go directly to the Pacing Guide and Timeline Overview\)](#)

Bridging Instructional Vision & Practice in Mathematics in Lynn Public Schools

Dear Educators,

Welcome back to SY25-26! As we begin this school year, I want to pause and center us on the vision behind our curriculum maps. These maps are not just pacing tools; they are a **shared blueprint** for how we **create language-rich, rigorous, and accessible math classrooms for every student**.

In everyday conversation, rigor often gets confused with “harder work.” In mathematics, rigor means something very different. True **rigor** lives in the **balance of three components**:

- **Conceptual Understanding:** Students understand the “why” of mathematics and can make connections across ideas.
- **Procedural Skill and Fluency:** Students can carry out calculations accurately, efficiently, and flexibly.
- **Application:** Students can apply mathematics to solve problems in real-world and novel contexts.

Our curriculum maps and professional learning are designed to support you in weaving these three dimensions together, always with an eye on the language demands of math. We know students need **intentional opportunities to speak, write, listen, and reason in mathematics**. By embedding **clear objectives, scaffolds, and Math Language Routines (MLRs)**, we ensure every learner has access to **rigorous mathematics and every student’s voice is heard**.

As you engage with these maps, use them not only as a planning tool but as a guide to instruction that is **equitable, language-rich, and standards-aligned**. The work you do each day, **facilitating discourse, modeling thinking, and holding students to high expectations**, is what brings these maps to life.

Thank you for your commitment to our students and to this vision. Together, we will make mathematics classrooms places where all learners can thrive, grow in confidence, and see themselves as mathematicians.

With gratitude and partnership,

Kerri Gaudet

Assistant Director of Curriculum & Instruction, Secondary Mathematics

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Components of a high- quality math instruction:

Rigor (Full Standard)

- Conceptual understanding, procedural skill/fluency, and application
- Tasks require explanation, justification, and connections
- Multiple modes of representation

Math Language Routines (MLRs)

- Structured student-to-student discourse
- MLRs visible in practice
- All students, including multilingual learners, using academic language

Objectives

- Content and language objectives posted and referenced
- Students can articulate learning and success criteria

Scaffolds

- Purposeful supports without lowering cognitive demand
- Supports fade over time

Color Key:  Rigor |  MLRs |  Objectives |  Scaffolds

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Language Development

Language for Social and Instructional Purposes

- This language is across all disciplines and school settings.
 - Greetings
 - Establishing routines
 - Following procedures
 - Asking for clarification
 - Discussing with peers
 - Relating personal ideas, feelings, and views
 - Exploring languages, cultures, and perspectives

Students communicate to learn but also to convey personal needs and wants, to affirm their own identities, and to form and maintain relationships

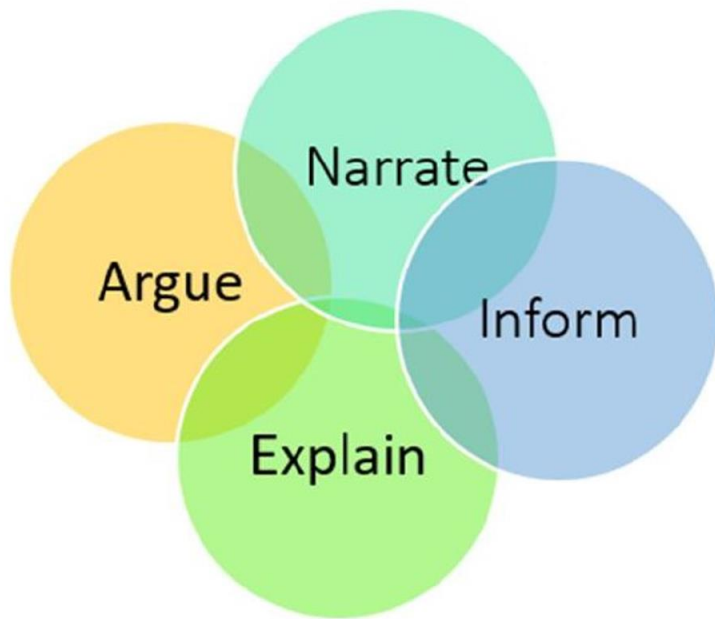
Language for Mathematics

- This is discipline specific language
- This is language that all students are likely building
- Cognizant of how mathematicians use language and helping students develop that control of disciplinary language
 - Example: $|-3|$ or 4^2
- What students say externally and hear spoken around them is what they will say in their internal self talk

From the [*WIDA 2020 Standards*](#) (p. 24-25)

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Language for What Purpose?



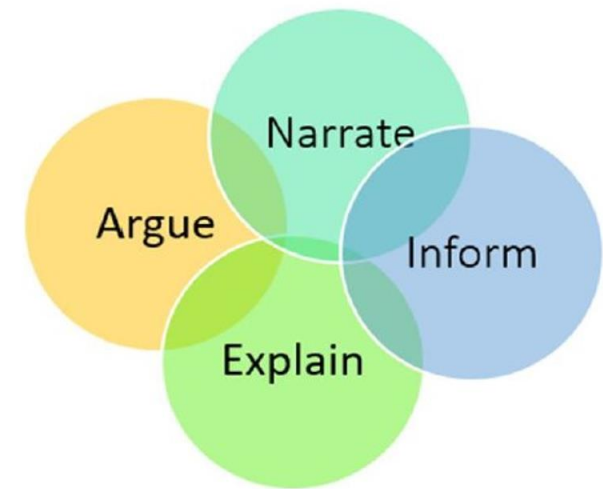
- Key Language Uses (KLUs) are ways students use language to accomplish content specific tasks.
- They support explicit integration of content and language instruction
- High leverage-portable language uses!

KLU = Key Language Use from the [WIDA 2020 Standards](#) (p. 26-27)

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KLUs predominately found in content Standards

Narrate	Inform	**Explain	*Argue
Describe	Tell	Clarify	Reason (apply)
Recount	Instruct	Show	Defend
Report		Determine	Conjecture
Repeat		Justify (apply)	Present (evidence)
State		Answer	Claim
		Illustrate (model)	Cite
		Interpret	Justify
		Solve	Refute
		Translate	Draw (conclusion)
		Simplify	
		Demonstrate	
		Tell	



*Argument answers the question: How do you know? This is a request for evidence.

**Explanation answers the question: Why is that so? This is a request for cause.

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Mathematical Language Routines Overview

Routine	Overview
1 Stronger and Clearer Each Time	Purpose: To provide a structured and interactive opportunity for students to revise and refine both their ideas and their verbal and written output (Zwiers, 2014). Example: Successive Pair Shares; Convince Yourself, a friend, a Skeptic; Writing Math Definitions,
2 Collect and Display	Purpose: To capture students' oral words and phrases into a stable, collective reference. Example: Gather & Show Student Discourse; Number Talks
3 Critique, Correct, and Clarify	Purpose: To give students a piece of mathematical writing that is not their own to analyze, reflect on, and develop. Example: Critique a Partial or Flawed Response; Always-Sometimes-Never
4 Information Gap	Purpose: To create a need for students to communicate (Gibbons, 2002). Example: Info Gap Cards; Info Gap Games
5 Co-Craft Questions & Problems	Purpose: To allow students to get inside of a context before feeling pressure to produce answers, to create space for students to produce the language of mathematical questions themselves, and to provide opportunities for students to analyze how different mathematical forms can represent different situations. Example: Co-Craft Questions; Co-Craft Problems; Co-Craft Situations, Numberless Graphs/Word Problems
6 Three Reads	Purpose: To ensure that students know what they are being asked to do, create opportunities for students to reflect on the ways mathematical questions are presented, and equip students with tools used to negotiate meaning (Kelemanik, Lucenta & Creighton, 2016). Example: Three Reads Routine; Values/Units Chart
7 Compare and Connect	Purpose: To foster students' meta-awareness as they identify, compare, and contrast different mathematical approaches, representations, concepts, examples, and language. Example: Which one Doesn't Belong? Same/Different Stand & Talk.
8 Discussion Supports	Purpose: To support rich and inclusive discussions about mathematical ideas, representations, contexts, and strategies (Chapin, O'Connor, & Anderson, 2009) Example: Whole Class Discussion Supports; Numbered Heads Together;

Algebra 2 Timeline Overview

<i>Topic</i>	<i>Approx. # Hours</i>	<i>Approx. # Weeks</i>	<i>Assessments</i>	<i>Assessment Dates</i>
Prerequisite, Procedures, Routines	8	1.5	N/A	
	1-2		Progress Monitoring Form A	9/13
Topic 1: Linear Functions and Systems	13	2.5	Topic 1 – Topic Assessment A	10/2
Topic 2: Quadratic Functions and Equations	17	3.5	Topic 2 - Topic Assessment A	10/25
Topic 3: Polynomial Functions	15	3	Topic 3 - Topic Assessment A	11/15
Topic 4: Rational Functions	13	2.5	Topic 4 - Topic Assessment A	12/6
Topic 5: Rational Exponents and Radical Functions	18	3.5	Topic 5 - Topic Assessment A	1/10
Topic 6: Exponential and Logarithmic Functions	13	2.5	Topic 6 - Topic Assessment A	1/31
	1-2		Progress Monitoring Form B	2/3
Topic 7: Trigonometric Functions	10	2	Topic 7 - Topic Assessment A	2/14
Topic 8: Trigonometric Equations and Identities	10	2	Topic 8 - Topic Assessment A	3/7
Topic 9: Probability (Savvas Topic 12)	15	3	Topic 12 - Topic Assessment A	3/28
Topic 10: Conic Sections (Savvas Topic 9)	10	2	Topic 9 - Topic Assessment A	4/17
Topic 11: Matrices (Savvas Topic 10)	15	3	Topic 10 - Topic Assessment A	5/16
	1-2		Progress Monitoring Form C	5/19
Topic 12: Data Analysis and Statistics (Savvas Topic 11)	15	3	Topic 11 - Topic Assessment A	6/11
	175-178	34		

Topic 1: Linear Functions and Systems

Duration: 13 days (2.5 weeks)

[\(Back to timeline overview\)](#)

Assessment Date: 10/2

Focus of Topic 1

Topic 1 focuses on extending students' previous knowledge of functions. Students identify key features of functions and understand how to interpret graphs of functions. Students learn methods for solving equations and inequalities and systems of linear equations and inequalities by using graphing, tables, and matrices.

Detailed description of Focus

Math Background: Coherence – “Looking Back, In this Topic, Looking Ahead”

Math Background: Rigor – Conceptual understanding, Procedural Skill and Fluency, Applications

MA Curriculum Standards:

The order of the standards shown below does not imply the sequence in which they should be taught.

HS-A High School Algebra

CED Creating Equations

A. Create equations that describe numbers or relationships.

1. Create equations and inequalities in one variable and use them to solve problems. (Include equations arising from linear and quadratic functions, and simple root and rational functions and exponential functions.)*
2. Create equations in two or more variables to represent relationships between quantities; graph equations on coordinate axes with labels and scales.*
3. Represent constraints by equations or inequalities, and by systems of equations and/or inequalities, and interpret solutions as viable or non-viable options in a modeling context.* *For example, represent inequalities describing nutritional and cost constraints on combinations of different foods.*

REI Reasoning with Equations and Inequalities

C. Solve systems of equations.

6. Solve systems of linear equations exactly and approximately (e.g., with graphs), focusing on pairs of linear equations in two variables.

D. Represent and solve equations and inequalities graphically.

11. Explain why the x-coordinates of the points where the graphs of the equations $y = f(x)$ and $y = g(x)$ intersect are the solutions of the equation $f(x) = g(x)$; find the solutions approximately, e.g., using technology to graph the functions, make tables of values, or find successive approximations. Include cases where $f(x)$ and/or $g(x)$ are linear, polynomial, rational, absolute value, exponential, and logarithmic functions.*

HS-F High School Functions

BF - Building Functions

A. Build a function that models a relationship between two quantities.

1. Write a function (linear, quadratic, exponential, simple rational, radical, logarithmic, and trigonometric) that describes a relationship between two quantities.*
 - a. Determine an explicit expression, a recursive process, or steps for calculation from a context.*
2. Write arithmetic and geometric sequences both recursively and with an explicit formula, use them to model situations, and translate between the two forms.*

B. Build new functions from existing functions.

3. Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. (Include linear, quadratic, exponential, absolute value, simple rational and radical, logarithmic and trigonometric functions.) Utilize technology to experiment with cases and illustrate an explanation of the effects on the graph. (Include recognizing even and odd functions from their graphs and algebraic expressions for them.)

IF - Interpreting Functions

A. Understand the concept of a function and use function notation.

3. Recognize that sequences are functions, sometimes defined recursively, whose domain is a subset of the integers. *For example, the Fibonacci sequence is defined recursively by $f(0) = f(1) = 1$, $f(n + 1) = f(n) + f(n - 1)$ for $n \geq 1$.*

B. Interpret functions that arise in applications in terms of the context (linear, quadratic, exponential, rational, polynomial, square root, cube root, trigonometric, logarithmic).

4. For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity. *

5. Relate the domain of a function to its graph and, where applicable, to the quantitative relationship it describes.* *For example, if the function $h(n)$ gives the number of person-hours it takes to assemble n engines in a factory, then the positive integers would be an appropriate domain for the function.*

6. Calculate and interpret the average rate of change of a function (presented symbolically or as a table) over a specified interval. Estimate the rate of change from a graph.*

C. Analyze functions using different representations.

7. Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases.*

b. Graph square root, cube root, and piecewise-defined functions, including step functions and absolute value functions.*

LE - Linear, Quadratic, and Exponential Models

A. Construct and compare linear, quadratic, and exponential models and solve problems.

2. Construct linear and exponential functions, including arithmetic and geometric sequences, given a graph, a description of a relationship, or two input-output pairs (including reading these from a table).*

[MCAS Digital Item Library](#)– Please navigate to the appropriate standard and ensure the selected questions align with your lesson focus

Highlighted Math Practices for this Topic:

MP.3 Construct viable arguments
MP.7 Look for and make sure of structure

[Details of the above practices](#)

Topic Essential Question:

What are the ways in which functions can be used to represent and solve problems involving quantities?

Objectives / Student Outcomes:

[Topic Planner Lessons 1 through 4](#)

[Lesson 1-1](#) Key Features of Functions [Materials](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

[Suggested Supplemental](#)

- ✓ Identify key features of a graph of a function, including the intercepts, positive and negative intervals, and areas where the function is increasing or decreasing.
- ✓ Calculate and interpret the average rate of change of a function over a specified interval.

- ✓ Write the domain and range of functions using set-builder and interval notations.

Lesson 1-2 Transformations of Functions

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Graph a transformed function by identifying the effect of on the graph of replacing $f(x)$ by $f(x)+k$, $kf(x)$, $f(kx)$ and $f(x+k)$ for specific values of k .
- ✓ Write an equation of a transformed function.
- ✓ Relate the domain of a function to its graph and the real-world situation it describes

Lesson 1-3 Piecewise-Defined Functions

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

[Supplemental Materials](#)

- ✓ Create and graph piecewise-defined functions, including absolute value functions and step functions.
- ✓ Create and use a piecewise-defined function from real-world data.
- ✓ Write a piecewise-defined rule from a graph.

Lesson 1-4 Arithmetic Sequences and Series

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Identify the common difference in an arithmetic sequence.
- ✓ Write arithmetic sequences both recursively and with an explicit formula.
- ✓ Construct arithmetic sequences, given a graph, a description of a relationship, or two input-output pairs.

Topic Planner Lessons 5 through 6

Lesson 1-5 Solving Equations and Inequalities by Graphing

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Use graphs, tables, and graphing technology to find or approximate solutions to equations and inequalities.
- ✓ Find approximate solutions to equations and inequalities by setting each expression equal to y and graphing.

Lesson 1-6 Linear Systems

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Solve linear systems graphically and algebraically.
- ✓ Identify regions that satisfy systems of inequalities.

Primary Resource:

[TE – SAVVAS enVision Algebra 2 Topic 1](#)

Topic 2: Quadratic Functions and Equations

Duration: 17 days hours (3.5 weeks)

[\(Back to timeline overview\)](#)

Assessment Date: 10/25

Focus of Topic 2

Topic 2 focuses on extending previous understanding of quadratic functions. Students identify different forms of quadratic functions and their key features. Students explore complex numbers and solve problems with complex numbers. Students learn different methods for solving quadratic equations.

Detailed description of Focus

Math Background: Coherence - “Looking Back, In this Topic, Looking Ahead”

Math Background: Rigor – Conceptual understanding, Procedural Skill and Fluency, Applications

MA Curriculum Standards:

The order of the standards shown below does not imply the sequence in which they should be taught.

HS-A High School Algebra

APR - Arithmetic with Polynomials and Rational Expressions

B. Understand the relationship between zeros and factors of polynomials.

3. Identify zeros of polynomials when suitable factorizations are available, and use the zeros to construct a rough graph of the function defined by the polynomial.

C. Use polynomial identities to solve problems.

4. Prove polynomial identities and use them to describe numerical relationships. *For example, the polynomial identity*

$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ *can be used to generate Pythagorean triples.*

CED - Creating Equations

A. Create equations that describe numbers or relationships.

2. Create equations in two or more variables to represent relationships between quantities; graph equations on coordinate axes with labels and scales.*

SSE - Seeing Structure in Expressions

A. Interpret the structure of linear, quadratic, exponential, polynomial, and rational expressions.

2. Use the structure of an expression to identify ways to rewrite it. *For example, see $x^4 - y^4$ as $(x^2)^2 - (y^2)^2$, thus recognizing it as a difference of squares that can be factored as $(x^2 - y^2)(x^2 + y^2)$.*

B. Write expressions in equivalent forms to solve problems.

3. Choose and produce an equivalent form of an expression to reveal and explain properties of the quantity represented by the expression.

a. Factor a quadratic expression to reveal the zeros of the function it defines.

REI - Reasoning with Equations and Inequalities

A. Understand solving equations as a process of reasoning and explain the reasoning.

1. Explain each step in solving a simple equation as following from the equality of numbers asserted at the previous step, starting from the assumption that the original equation has a solution. Construct a viable argument to justify or refute a solution method.

B. Solve equations and inequalities in one variable.

4. Solve quadratic equations in one variable.

a. Use the method of completing the square to transform any quadratic equation in x into an equation of the form $(x - p)^2 = q$ that has the same solutions. Derive the quadratic formula from this form.

b. Solve quadratic equations by inspection (e.g., for $x^2 = 49$), taking square roots, completing the square, the quadratic formula, and factoring, as appropriate to the initial form of the equation. Recognize when the quadratic formula gives complex solutions and write them as $a \pm bi$ for real numbers a and b .

C. Solve systems of equations.

7. Solve a simple system consisting of a linear equation and a quadratic equation in two variables algebraically and graphically. *For example, find the points of intersection between the line $y = -3x$ and the circle $x^2 + y^2 = 3$.*

D. Represent and solve equations and inequalities graphically.

11. Explain why the x -coordinates of the points where the graphs of the equations $y = f(x)$ and $y = g(x)$ intersect are the solutions of the equation $f(x) = g(x)$; find the solutions approximately, e.g., using technology to graph the functions, make tables of values, or find successive approximations. Include cases where $f(x)$ and/or $g(x)$ are linear, polynomial, rational, absolute value, exponential, and logarithmic functions.*

12. Graph the solutions of a linear inequality in two variables as a half-plane (excluding the boundary in the case of a strict inequality), and graph the solution set of a system of linear inequalities in two variables as the intersection of the corresponding half-planes.

HS-F High School Functions

IF - Interpreting Functions

B. Interpret functions that arise in applications in terms of the context (linear, quadratic, exponential, rational, polynomial, square root, cube root, trigonometric, logarithmic).

4. For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity.*

C. Analyze functions using different representations.

7. Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases.*

a. Graph linear and quadratic functions and show intercepts, maxima, and minima.*

BF - Building Functions

B. Build new functions from existing functions.

3. Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. (Include linear, quadratic, exponential, absolute value, simple rational and radical, logarithmic and trigonometric functions.) Utilize technology to experiment with cases and illustrate an explanation of the effects on the graph. (Include recognizing even and odd functions from their graphs and algebraic expressions for them.)

HS-N High School Number and Quantity

CN - The Complex Number System

A. Perform arithmetic operations with complex numbers.

1. Know there is a complex number i such that $i^2 = -1$, and every complex number has the form $a + bi$ with a and b real.

2. Use the relation $i^2 = -1$ and the Commutative, Associative, and Distributive properties to add, subtract, and multiply complex numbers

3. (+) Find the conjugate of a complex number; use conjugates to find moduli and quotients of complex numbers.

C. Use complex numbers in polynomial identities and equations.

7. Solve quadratic equations with real coefficients that have complex solutions.

HS-S High School Statistics and Probability

ID Interpreting Categorical and Quantitative Data

B. Summarize, represent, and interpret data on two categorical and quantitative variables.

6. Represent data on two quantitative variables on a scatter plot, and describe how the variables are related.*

- a. Fit a linear function to the data and use the fitted function to solve problems in the context of the data. Use functions fitted to data or choose a function suggested by the context. Emphasize linear and exponential models.*

[MCAS Digital Item Library](#)– Please navigate to the appropriate standard and ensure the selected questions align with your lesson focus

Highlighted Math Practices for this Topic:

MP.1 - Make Sense of Problems and Persevere in Solving Them
MP.7 – Look for and Make Use of Structure

[Details of the above practices](#)

Topic Essential Question:

How do you use quadratic functions to model situations and solve problems?

Objectives / Student Outcomes:

[Topic Planner Lessons 1 through 4](#)

[Lesson 2-1 Vertex Form of a Quadratic Function](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Create quadratic functions in vertex form to represent relationships between variables as shown in their graphs.
- ✓ Graph functions on coordinate axes using their key features.
- ✓ Interpret key features of the graph of a quadratic function.

[Lesson 2-2 Standard Form of a Quadratic Equation](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Create quadratic functions written in standard form.
- ✓ Identify key features of quadratic functions and graph a quadratic function written in standard form.

[Lesson 2-3 Factored Form of a Quadratic Function](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Write a quadratic equation in factored form and use it to identify the zeros of the function it defines.
- ✓ Determine the intervals over which a quadratic function is positive or negative.

[Lesson 2-4 Complex Numbers and Operations](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Add, subtract, and multiply complex numbers using the properties of operations and the relation $i^2=-1$
- ✓ Use complex numbers to represent numbers that are not on the real number line.

[Topic Planner Lessons 5 through 7](#)

[Lesson 2-5 Completing the Square](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Transform a quadratic equation into the form $(x-p)^2 = q$ by completing the square.
- ✓ Complete the square to reveal the minimum or maximum value of a quadratic expression.

[Lesson 2-6 The Quadratic Formula](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

[Supplementary Materials](#)

- ✓ Use the Quadratic Formula to solve quadratic equations that have complex solutions.

[Lesson 2-7 Linear-Quadratic Systems](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Use algebra to solve a linear-quadratic system.
- ✓ Solve a linear-quadratic system using graphing and explain why the points of intersection are the solutions.

Primary Resource:

[TE – SAVVAS enVision Algebra 2 Topic 2](#)

Topic 3: Polynomial Functions

Duration: 15 days (3 weeks)

[\(Back to timeline overview\)](#)

Assessment Date: 11/15

Focus of Topic 2

Topic 3 focuses on extending students' previous knowledge of polynomials. Students identify the key features of polynomial functions and interpret graphs of polynomial functions. They learn methods to add, subtract, multiply, and divide polynomial expressions. They use polynomial identities to multiply and factor polynomial expressions, use multiple theorems as tools to understand the roots of polynomial functions, and transform graphs from cubic or quartic parent functions.

Detailed description of Focus

Math Background: Coherence - "Looking Back, In this Topic, Looking Ahead"

Math Background: Rigor – Conceptual understanding, Procedural Skill and Fluency, Applications

MA Curriculum Standards:

The order of the standards shown below does not imply the sequence in which they should be taught.

HS-A High School Algebra

APR - Arithmetic with Polynomials and Rational Expressions

A. Perform arithmetic operations on polynomials.

1. Understand that polynomials form a system analogous to the integers, namely, they are closed under certain operations.

B. Understand the relationship between zeros and factors of polynomials.

2. Know and apply the Remainder Theorem: For a polynomial $p(x)$ and a number a , the remainder on division by $x - a$ is $p(a)$, so $p(a) = 0$ if and only if $(x - a)$ is a factor of $p(x)$.

3. Identify zeros of polynomials when suitable factorizations are available, and use the zeros to construct a rough graph of the function defined by the polynomial.

C. Use polynomial identities to solve problems.

4. Prove polynomial identities and use them to describe numerical relationships. *For example, the polynomial identity*

$$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2 \text{ can be used to generate Pythagorean triples.}$$

D. Rewrite rational expressions.

6. Rewrite simple rational expressions in different forms; write in the form $\frac{a(x)}{b(x)} = q(x) + \frac{r(x)}{b(x)}$ where $a(x)$, $b(x)$, $q(x)$, and $r(x)$ are polynomials with the degree of $r(x)$ less than the degree of $b(x)$, using inspection, long division, or, for the more complicated examples, a computer algebra system.

BF - Building Functions

A. Build a function that models a relationship between two quantities.

1. Write a function (linear, quadratic, exponential, simple rational, radical, logarithmic, and trigonometric) that describes a relationship between two quantities.*

b. Combine standard function types using arithmetic operations.*

B. Build new functions from existing functions.

3. Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. (Include linear, quadratic, exponential, absolute value, simple rational and radical, logarithmic and trigonometric functions.) Utilize technology to experiment with cases and illustrate an explanation of the effects on the graph. (Include recognizing even and odd functions from their graphs and algebraic expressions for them.)

SSE - Seeing Structure in Expressions

A. Interpret the structure of linear, quadratic, exponential, polynomial, and rational expressions.

1. Interpret expressions that represent a quantity in terms of its context.*
 - a. Interpret parts of an expression, such as terms, factors, and coefficients
2. Use the structure of an expression to identify ways to rewrite it. *For example, see $x^4 - y^4$ as $(x^2)^2 - (y^2)^2$, thus recognizing it as a difference of squares that can be factored as $(x^2 - y^2)(x^2 + y^2)$.*

HS-F High School Functions

IF - Interpreting Functions

B. Interpret functions that arise in applications in terms of the context (linear, quadratic, exponential, rational, polynomial, square root, cube root, trigonometric, logarithmic).

4. For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity.*
6. Calculate and interpret the average rate of change of a function (presented symbolically or as a table) over a specified interval. Estimate the rate of change from a graph.*

C. Analyze functions using different representations.

7. Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases.*
 - c. Graph polynomial functions, identifying zeros when suitable factorizations are available, and showing end behavior.*
9. Translate among different representations of functions (algebraically, graphically, numerically in tables, or by verbal descriptions). Compare properties of two functions each represented in a different way. *For example, given a graph of one polynomial function (including quadratic functions) and an algebraic expression for another, say which has the larger/smaller relative maximum and/or minimum.*

HS-N High School Number and Quantity

CN - The Complex Number System

C. Use complex numbers in polynomial identities and equations.

8. (+) Extend polynomial identities to the complex numbers. For example, rewrite $x^2 + 4$ as $(x + 2i)(x - 2i)$.
9. (+) Know the Fundamental Theorem of Algebra; show that it is true for quadratic polynomials.

[MCAS Digital Item Library](#)– Please navigate to the appropriate standard and ensure the selected questions align with your lesson focus

Highlighted Math Practices for this Topic:

MP.1 - Make Sense of Problems and Persevere in Solving Them
MP.7 – Look for and Make Use of Structure

[Details of the above practices](#)

Topic Essential Question:

What can an equation for a polynomial function tell about its graph, and what can a graph of a polynomial function tell about the solutions of a polynomial equation?

Objectives / Student Outcomes:

[Topic Planner Lessons 1 through 4](#)

[Lesson 3-1 Graphing Polynomial Functions](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Graph polynomial functions and show the key features of the graph.

- ✓ Predict the end behavior of polynomial functions by interpreting the leading coefficients and degrees.
- ✓ Sketch graphs showing key features, given a verbal description.

Lesson 3-2 Adding, Subtracting, and Multiplying Polynomials

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Add, subtract, and multiply polynomials and understand that polynomials are closed under these operations.
- ✓ Compare a polynomial function represented algebraically with one represented graphically.

Lesson 3-3 Polynomial Identities

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Prove polynomial identities and use them to multiply and factor polynomials.
- ✓ Expand binomials using the Binomial Theorem and coefficients determined by Pascal's Triangle.
- ✓ Determine the intervals over which a quadratic function is positive or negative.

Lesson 3-4 Dividing Polynomials

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Divide polynomial expressions using long division.
- ✓ Use synthetic division to rewrite rational expressions.

Topic Planner Lessons 5 through 7

Lesson 3-5 Zeros of Polynomial Functions

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Identify the zeros of a function by factoring or using synthetic division.
- ✓ Use the zeros of a polynomial functions to sketch its graph.

Lesson 3-6 Theorems About Roots of Polynomial Equations

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Extend polynomial theorems and identities to find the real and complex solutions of a polynomial equation.
- ✓ Write polynomial functions using conjugates.

Lesson 3-7 Transformations of Polynomial Functions

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Recognize even and odd functions from their graphs and algebraic equations.
- ✓ Identify the effect on the graphs of cubic and quartic functions of replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$.
- ✓ Given a graph, determine the equation as it is related to its parent cubic function or quartic function.

Primary Resource:

[TE – SAVVAS enVision Algebra 2 Topic 3](#)

Topic 4: Rational Functions

Duration: 13 days (2.5 weeks)

[\(Back to timeline overview\)](#)

Assessment Date: 12/6

Focus of Topic 4

Topic 4 focuses on extending students' previous knowledge of polynomial functions to rational functions. Students identify the key features of the graphs of rational functions. Students learn methods of solving rational equations.

Detailed description of Focus

Math Background: Coherence – “Looking Back, In this Topic, Looking Ahead”

Math Background: Rigor – Conceptual understanding, Procedural Skill and Fluency, Applications

MA Curriculum Standards:

The order of the standards shown below does not imply the sequence in which they should be taught.

HS-A High School Algebra

CED - Creating Equations

A. Create equations that describe numbers or relationships.

1. Create equations and inequalities in one variable and use them to solve problems. (Include equations arising from linear and quadratic functions, and simple root and rational functions and exponential functions.)«
2. Create equations in two or more variables to represent relationships between quantities; graph equations on coordinate axes with labels and scales. *

APR - Arithmetic and Polynomials and Rational Expressions

D. Rewrite rational expressions.

6. Rewrite simple rational expressions in different forms; write $a(x)/b(x)$ in the form $q(x) + r(x)/b(x)$, where $a(x)$, $b(x)$, $q(x)$, and $r(x)$ are polynomials with the degree of $r(x)$ less than the degree of $b(x)$, using inspection, long division, or, for the more complicated examples, a computer algebra system.
7. Understand that rational expressions form a system analogous to the rational numbers, closed under addition, subtraction, multiplication, and division by a nonzero rational expression; add, subtract, multiply, and divide rational expressions. (+)

REI - Reasoning with Equations and Inequalities

A. Understand solving equations as a process of reasoning and explain the reasoning.

1. Explain each step in solving a simple equation as following from the equality of numbers asserted at the previous step, starting from the assumption that the original equation has a solution. Construct a viable argument to justify or refute a solution method.

D. Represent and solve equations and inequalities graphically.

11. Explain why the x-coordinates of the points where the graphs of the equations $y = f(x)$ and $y = g(x)$ intersect are the solutions of the equation $f(x) = g(x)$; find the solutions approximately, e.g., using technology to graph the functions, make tables of values, or find successive approximations. Include cases where $f(x)$ and/or $g(x)$ are linear, polynomial, rational, absolute value, exponential, and logarithmic functions.*

SSE - Seeing Structure in Expressions

A. Interpret the structure of linear, quadratic, exponential, polynomial, and rational expressions.

1. Interpret expressions that represent a quantity in terms of its context. * For example, interpret $P(1 + r)^n$ as the product of P and a factor not depending on P .
 - a. Interpret parts of an expression, such as terms, factors, and coefficients.
 - b. Interpret complicated expressions by viewing one or more of their parts as a single entity.

2. Use the structure of an expression to identify ways to rewrite it. *For example, see $x^4 - y^4$ as $(x^2)^2 - (y^2)^2$, thus recognizing it as a difference of squares that can be factored as $(x^2 - y^2)(x^2 + y^2)$.*

HS-F High School Functions

BF - Building Functions

A. Understand the concept of a function and use function notation.

1. Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then $f(x)$ denotes the output of f corresponding to the input x . The graph of f is the graph of the equation $y = f(x)$.

B. Build new functions from existing functions.

3. Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. (Include linear, quadratic, exponential, absolute value, simple rational and radical, logarithmic and trigonometric functions.) Utilize technology to experiment with cases and illustrate an explanation of the effects on the graph. (Include recognizing even and odd functions from their graphs and algebraic expressions for them.)

IF - Interpret Functions

B. Interpret functions that arise in applications in terms of the context (linear, quadratic, exponential, rational, polynomial, square root, cube root, trigonometric, logarithmic).

4. For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity.*

C. Analyze functions using different representations.

7. Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases. *

d. Graph rational functions, identifying zeros and asymptotes when suitable factorizations are available, and showing end behavior.(+)

[MCAS Digital Item Library](#)– Please navigate to the appropriate standard and ensure the selected questions align with your lesson focus

Highlighted Math Practices for this Topic:

MP.1 Make sense of problems and persevere in solving them
MP.6 Attend to Precision

[Details of the above practices](#)

Topic Essential Question:

What are rational functions, and what are the key features of their graphs?

Objectives / Student Outcomes:

[Topic Planner Lessons 1 through 3](#)

[Lesson 4-1 Inverse Variation and the Reciprocal Function](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Use inverse variation to write and graph the reciprocal function.
- ✓ Identify the effect of transformations on the graph of the reciprocal function and define the effects of h and k on the function $f(x) = \frac{1}{x-h} + k$.

[Lesson 4-2 Graphing Rational Functions](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Graph rational functions by identifying asymptotes and end behavior.
- ✓ Rewrite simple rational expressions in different forms using long division.

[Lesson 4-3 Multiplying and Dividing Rational Expressions](#)

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Use the structure of rational expressions to rewrite simple rational expressions in different forms.

- ✓ Understand that rational expressions form a system analogous to the system of rational numbers and use that understanding to multiply and divide rational expressions.

Topic Planner Lessons 4 through 5

Lesson 4-4 Adding and Subtracting Rational Expressions

(Lesson Presentation)

(Lesson Plan)

- ✓ Understand that rational expressions form a system analogous to the system of rational numbers and use that understanding to add and subtract rational expressions.

Lesson 4-5 Solving Rational Equations

(Lesson Presentation)

(Lesson Plan)

- ✓ Solve rational equations in one variable.
- ✓ Identify extraneous solutions to rational equations and give examples of how they arise.

Primary Resource:

TE – SAVVAS enVision Algebra 2 Topic 4

Topic 5: Rational Exponents and Radical Functions

Duration: 18 days (3.5 weeks)

[\(Back to timeline overview\)](#)

Assessment Date: 1/10

Focus of Topic 5

Topic 5 extends knowledge of radical functions. Students understand properties of rational exponents and radicals. They learn methods to graph radical functions, solve radical equations, and combine functions. Students identify inverses of functions and learn to write the equations of inverse functions.

Detailed description of Focus

Math Background: Coherence – “Looking Back, In this Topic, Looking Ahead”

Math Background: Rigor – Conceptual understanding, Procedural Skill and Fluency, Applications

MA Curriculum Standards:

The order of the standards shown below does not imply the sequence in which they should be taught.

HS-N Number and Quantity Content

N-RN The Real Number System

A. Extend the properties of exponents to rational exponents

1. Explain how the definition of the meaning of rational exponents follows from extending the properties of integer exponents to those values, allowing for a notation for radicals in terms of rational exponents. *For example, we define $5^{1/3}$ to be the cube root of 5 because we want $(5^{1/3})^3 = 5$ $(1/3)^3$ to hold, so $(5^{1/3})^3$ must equal 5.*
2. Rewrite expressions involving radicals and rational exponents using the properties of exponents.

HS-A High School Algebra

A-CED Creating Equations

A. Create equations that describe numbers or relationships.

1. Create equations and inequalities in one variable and use them to solve problems. Include equations arising from linear, and quadratic, functions and simple rational and exponential functions with integer exponents.
3. Represent constraints by linear equations or inequalities, and by systems of linear equations and/or inequalities and interpret solutions as viable or non-viable options in a modeling context.
4. Rearrange formulas to highlight a quantity of interest using the same reasoning as in solving equations (Properties of Equality). For example, rearrange Ohm’s law $R=V^2/P$ to solve to highlight for voltage V . Manipulate variables in formulas used in financial contexts such as for simple interest, $(I = Prt)$.

A-REI Reasoning with Equations and Inequalities

A. Understand solving equations as a process of reasoning and explain the reasoning.

1. Explain each step in solving a simple equation as following from the equality of numbers asserted at the previous step, starting from the assumption that the original equation has a solution. Construct a viable argument to justify or refute a solution method.
2. Solve simple rational and radical equations in one variable, and give examples showing how extraneous solutions may arise.

A-SSE Seeing Structure in Expressions

A. Interpret the structure of linear, quadratic, exponential, polynomial, and rational expressions.

1. Interpret expressions that represent a quantity in terms of its context. * *For example, interpret $P(1 + r)^n$ as the product of P and a factor not depending on P .*

a. Interpret parts of an expression, such as terms, factors, and coefficients.

b. Interpret complicated expressions by viewing one or more of their parts as a single entity.

2. Use the structure of an expression to identify ways to rewrite it. *For example, see $x^4 - y^4$ as $(x^2)^2 - (y^2)^2$, thus recognizing it as a difference of squares that can be factored as $(x^2 - y^2)(x^2 + y^2)$.*

HS-F High School Functions

F-BF Building Functions

A. Understand the concept of a function and use function notation.

1. Write a function (linear, quadratic, exponential, simple rational, radical, logarithmic, and trigonometric) that describes a relationship between two quantities.*

b. Combine standard function types using arithmetic operations.* *For example, build a function that models the temperature of a cooling body by adding a constant function to a decaying exponential, and relate these functions to the model.*

c. Compose functions. * (+) *For example, if $T(y)$ is the temperature in the atmosphere as a function of height, and $h(t)$ is the height of a weather balloon as a function of time, then $T(h(t))$ is the temperature at the location of the weather balloon as a function of time.*

2. Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then $f(x)$ denotes the output of f corresponding to the input x . The graph of f is the graph of the equation $y = f(x)$.

B. Build new functions from existing functions.

3. Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. (Include linear, quadratic, exponential, absolute value, simple rational and radical, logarithmic and trigonometric functions.) Utilize technology to experiment with cases and illustrate an explanation of the effects on the graph. (Include recognizing even and odd functions from their graphs and algebraic expressions for them.)

4. Find inverse functions algebraically and graphically.

a. Solve an equation of the form $f(x) = c$ for a simple function f that has an inverse and write an expression for the inverse. (Include linear and simple polynomial, rational, and exponential functions.)

b. Verify by composition that one function is the inverse of another. (+)

c. Read values of an inverse function from a graph or a table, given that the function has an inverse. (+)

d. Produce an invertible function from a non-invertible function by restricting the domain. (+)

F-IF Interpret Functions

B. Interpret functions that arise in applications in terms of the context (linear, quadratic, exponential, rational, polynomial, square root, cube root, trigonometric, logarithmic).

4. For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity.*

6. Calculate and interpret the average rate of change of a function (presented symbolically or as a table) over a specified interval. Estimate the rate of change from a graph.*

C. Analyze functions using different representations.

7. Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases. *

b. Graph square root, cube root, and piecewise-defined functions, including step functions and absolute value functions.*

[MCAS Digital Item Library](#)– Please navigate to the appropriate standard and ensure the selected questions align with your lesson focus

Highlighted Math Practices for this Topic:

MP.2 Reason Abstractly and Quantitatively
MP.4 Model with Mathematics

[Details of the above practices](#)

Topic Essential Question:

How are rational exponents and radical equations used to solve real-world problems?

Objectives / Student Outcomes:

[Topic Planner Lessons 1 through 4](#)

[Lesson 5-1](#) *n*th Roots, Radicals, and Rational Exponents

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Find all real *n*th roots of a number
- ✓ Evaluate expressions with rational exponents
- ✓ Use *n*th roots to solve equations by rewriting expressions using the properties of exponents

[Lesson 5-2](#) Properties of Exponents and Radicals

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Use properties of exponents and radicals to identify ways to rewrite radical expressions
- ✓ Interpret radical expressions that represent a quantity in terms of its context

[Lesson 5-3](#) Graphing Radical Functions

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Graph radical functions, including square root and cube root functions
- ✓ Identify the effect of transformations on the key features of the graphs of radical functions

[Lesson 5-4](#) Solving Radical Functions

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Solve radical equations in one variable
- ✓ Explain how extraneous solutions may arise when solving radical equations
- ✓ Solve radical inequalities and apply the solution within a real-world context

[Topic Planner Lessons 5 through 6](#)

[Lesson 5-5](#) Function Operations

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Combine functions by addition, subtraction, multiplication, or division, and identify the domain of the result
- ✓ Compose functions, specifying the order in which the functions are applied and describing the domain of the composite function

[Lesson 5-6](#) Inverse Relations and Functions

[\(Lesson Presentation\)](#)

[\(Lesson Plan\)](#)

- ✓ Use tables, graphs, and equations to represent the inverse of a relation
- ✓ Write an equation for the inverse of a function by restricting the domain
- ✓ Verify that one function is the inverse of another using composition

Primary Resource:

[TE – SAVVAS enVision Algebra 2 Topic 5](#)

Topic 6: Exponential and Logarithmic Functions

Duration: 13 days (2.5 weeks)

[\(Back to timeline overview\)](#)

Assessment Date: 1/31

Focus of Topic 6

Topic 6 focuses on extending previous understanding of exponential functions. Students identify key features of exponential functions. Students understand logarithms and their properties. Students learn how to solve exponential and logarithmic equations.

Detailed description of Focus

Math Background: Coherence – “Looking Back, In this Topic, Looking Ahead”

Math Background: Rigor – Conceptual understanding, Procedural Skill and Fluency, Applications

MA Curriculum Standards:

The order of the standards shown below does not imply the sequence in which they should be taught.

HS-F High School Functions

F-IF Interpret Functions

A. Understand the concept of a function and use function notation.

3. Recognize that sequences are functions, sometimes defined recursively, whose domain is a subset of the integers. *For example, the Fibonacci sequence is defined recursively by $f(0) = f(1) = 1$, $f(n + 1) = f(n) + f(n - 1)$ for $n \geq 1$.*

B. Interpret functions that arise in applications in terms of the context (linear, quadratic, exponential, rational, polynomial, square root, cube root, trigonometric, logarithmic).

4. For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity.*

5. Relate the domain of a function to its graph and, where applicable, to the quantitative relationship it describes.*

C. Analyze functions using different representations.

7. Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases. *

e. Graph exponential and logarithmic functions, showing intercepts and end behavior, and trigonometric functions, showing period, midline, and amplitude.*

8. Write a function defined by an expression in different but equivalent forms to reveal and explain different properties of the function.

b. Use the properties of exponents to interpret expressions for exponential functions. Apply to financial situations such as identifying appreciation and depreciation rate for the value of a house or car some time after its initial purchase.

9. Translate among different representations of functions (algebraically, graphically, numerically in tables, or by verbal descriptions). Compare properties of two functions each represented in a different way. *For example, given a graph of one polynomial function (including quadratic functions) and an algebraic expression for another, say which has the larger/smaller relative maximum and/or minimum*

F-BF Building Functions

A. Understand the concept of a function and use function notation.

1. Write a function (linear, quadratic, exponential, simple rational, radical, logarithmic, and trigonometric) that describes a relationship between two quantities.*

a. Determine an explicit expression, a recursive process, or steps for calculation from a context.*

2. Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then $f(x)$ denotes the output of f corresponding to the input x . The graph of f is the graph of the equation $y = f(x)$.

B. Build new functions from existing functions.

3. Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. (Include linear, quadratic, exponential, absolute value, simple rational and radical, logarithmic and trigonometric functions.) Utilize technology to experiment with cases and illustrate an explanation of the effects on the graph. (Include recognizing even and odd functions from their graphs and algebraic expressions for them.)

4. Find inverse functions algebraically and graphically.

a. Solve an equation of the form $f(x) = c$ for a simple function f that has an inverse and write an expression for the inverse. (Include linear and simple polynomial, rational, and exponential functions.)

c. Read values of an inverse function from a graph or a table, given that the function has an inverse. (+)

5. Understand the inverse relationship between exponents and logarithms and use this relationship to solve problems involving logarithms and exponents. (+)

F-LE Linear, Quadratic, and Exponential Models

A. Construct and compare linear, quadratic, and exponential models and solve problems.

2. Construct linear and exponential functions, including arithmetic and geometric sequences, given a graph, a description of a relationship, or two input-output pairs (including reading these from a table).*

4. For exponential models, express as a logarithm the solution to $ab^{ct} = d$ where a , c , and d are numbers and the base b is 2, 10, or e ; evaluate the logarithm using technology. *

B. Interpret expressions for functions in terms of the situation they model.

5. Interpret the parameters in a linear or exponential function (of the form $f(x) = b x + k$) in terms of a context.*

HS-A High School Algebra

A-SSE Seeing Structure in Expressions

A. Interpret the structure of linear, quadratic, exponential, polynomial, and rational expressions.

1. Interpret expressions that represent a quantity in terms of its context. * For example, interpret $P(1 + r)^n$ as the product of P and a factor not depending on P .

b. Interpret complicated expressions by viewing one or more of their parts as a single entity.

2. Use the structure of an expression to identify ways to rewrite it. *For example, see $x^4 - y^4$ as $(x^2)^2 - (y^2)^2$, thus recognizing it as a difference of squares that can be factored as $(x^2 - y^2)(x^2 + y^2)$.*

B. Write expressions in equivalent forms to solve problems.

3. Choose and produce an equivalent form of an expression to reveal and explain properties of the quantity represented by the expression

c. Use the properties of exponents to transform expressions for exponential functions.

4. Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems.* *For example, calculate mortgage payments.*

A-CED Creating Equations

A. Create equations that describe numbers or relationships.

1. Create equations and inequalities in one variable and use them to solve problems. Include equations arising from linear, and quadratic, functions and simple rational and exponential functions with integer exponents.

4. Rearrange formulas to highlight a quantity of interest using the same reasoning as in solving equations (Properties of Equality). *For example, rearrange Ohm's law $R=V^2/P$ to solve to highlight for voltage V . Manipulate variables in formulas used in financial contexts such as for simple interest, ($I = Prt$).*

A-REI Reasoning with Equations and Inequalities

A. Understand solving equations as a process of reasoning and explain the reasoning.

1. Explain each step in solving a simple equation as following from the equality of numbers asserted at the previous step, starting from the assumption that the original equation has a solution. Construct a viable argument to justify or refute a solution method.

HS-S Statistics and Probability

S-ID Interpreting Categorical and Quantitative Data

B. Summarize, represent, and interpret data on two categorical and quantitative variables.

6. Represent data on two quantitative variables on a scatter plot, and describe how the variables are related. *

a. Fit a linear function to the data and use the fitted function to solve problems in the context of the data. Use functions fitted to data or choose a function suggested by the context. Emphasize linear and exponential models.*

[MCAS Digital Item Library](#)– Please navigate to the appropriate standard and ensure the selected questions align with your lesson focus

Traditional Pathway: Model Algebra 2 - Critical Areas

Building on their work with linear, quadratic, and exponential functions, students extend their repertoire of functions to include logarithmic, polynomial, rational, and radical functions in the Model Algebra II course. Students work closely with the expressions that define the functions, are facile with algebraic manipulations of expressions, and continue to expand and hone their abilities to model situations and to solve equations, including solving quadratic equations over the set of complex numbers and solving exponential equations using the properties of logarithms.

For the high school Model Algebra II course, instructional time should focus on four critical areas: (1) relate arithmetic of rational expressions to arithmetic of rational numbers; (2) expand understandings of functions and graphing to include trigonometric functions; (3) synthesize and generalize functions and extend understanding of exponential functions to logarithmic functions; and (4) relate data display and summary statistics to probability and explore a variety of data collection methods.

1. A central theme of this Model Algebra II course is that the arithmetic of rational expressions is governed by the same rules as the arithmetic of rational numbers. Students explore the structural similarities between the system of polynomials and the system of integers. They draw on analogies between polynomial arithmetic and base-ten computation, focusing on properties of operations, particularly the distributive property. Connections are made between multiplication of polynomials with multiplication of multi-digit integers, and division of polynomials with long division of integers. Students identify zeros of polynomials, including complex zeros of quadratic polynomials, and make connections between zeros of polynomials and solutions of polynomial equations. The Fundamental Theorem of Algebra is examined.
2. Building on their previous work with functions and on their work with trigonometric ratios and circles in the Model Geometry course, students now use the coordinate plane to extend trigonometry to model periodic phenomena.
3. Students synthesize and generalize what they have learned about a variety of function families. They extend their work with exponential functions to include solving exponential equations with logarithms. They explore the effects of transformations on graphs of diverse functions, including functions arising in an application, in order to abstract the general principle that transformations on a graph always have the same effect regardless of the type of the underlying function. They identify appropriate types of functions to model a situation, they adjust parameters to improve the model, and they compare models by analyzing appropriateness of fit and making judgments about the domain over which a model is a good fit. The description of modeling as “*the process of choosing and using mathematics and statistics to analyze empirical situations, to understand them better, and to make decisions*” is at the heart of this Model Algebra II course. The narrative discussion and diagram of the modeling cycle should be considered when knowledge of functions, statistics, and geometry is applied in a modeling context.
4. Students see how the visual displays and summary statistics they learned in earlier grades relate to different types of data and to probability distributions. They identify different ways of collecting data—including sample surveys, experiments, and simulations—and the role that randomness and careful design play in the conclusions that can be drawn.

Standards for Mathematical Practice Grades 9–12

The Standards for Mathematical Practice describe expertise that mathematics educators at all levels should seek to develop in their students. The Standards for Mathematical Practice describe ways in which developing student practitioners of the discipline of mathematics engage with the subject matter as they grow in mathematical maturity and expertise throughout the elementary, middle, and high school years.

1. Make sense of problems and persevere in solving them.

Mathematically proficient high school students analyze givens, constraints, relationships, and goals. They make conjectures about the form and meaning of the solution and plan a solution pathway rather than simply jumping into a solution attempt. They consider analogous problems, and try special cases and simpler forms of the original problem in order to gain insight into its solution. They monitor and evaluate their progress and change course if necessary. High school students might, depending on the context of the problem, transform algebraic expressions or change the viewing window on their graphing calculator to get the information they need. Mathematically proficient students can explain correspondences between equations, verbal descriptions, tables, and graphs or draw diagrams of important features and relationships, graph and interpret representations of data, and search for regularity or trends. Mathematically proficient students check their answers to problems using different methods of solving, and they continually ask themselves, “Does this make sense?” They can understand the approaches of others to solving complex problems and identify correspondences between different approaches.

2. Reason abstractly and quantitatively.

Mathematically proficient high school students make sense of the quantities and their relationships in problem situations. Students bring two complementary abilities to bear on problems involving quantitative relationships: the ability to decontextualize—to abstract a given situation and represent it symbolically, and manipulate the representing symbols as if they have a life of their own, without necessarily attending to their referents—and the ability to contextualize, to pause as needed during the manipulation process in order to probe into the referents for the symbols involved. Students can write explanatory text that conveys their mathematical analyses and thinking, using relevant and sufficient facts, concrete details, quotations, and coherent development of ideas. Students can evaluate multiple sources of information presented in diverse formats (and media) to address a question or solve a problem. Quantitative reasoning entails habits of creating a coherent representation of the problem at hand; considering the units involved; attending to the meanings of quantities, not just how to compute them; and knowing and flexibly using different properties of operations and objects.

3. Construct viable arguments and critique the reasoning of others.

Mathematically proficient high school students understand and use stated assumptions, definitions, and previously established results in constructing verbal and written arguments. They make conjectures and build a logical progression of statements to explore the truth of their conjectures. They are able to analyze situations by breaking them into cases, and can recognize and use counterexamples and specific textual evidence to form their arguments. They justify their conclusions, communicate them to others, and respond to the arguments of others. They reason inductively about data, making plausible arguments that take into account the context from which the data arose. Mathematically proficient students are also able to compare the effectiveness of two plausible arguments, distinguish correct logic or reasoning from that which is flawed, and—if there is a flaw in an argument—explain what it is and why. They can construct formal arguments relevant to specific contexts and tasks. High school students learn to determine domains to which an argument applies. Students listen or read the arguments of others, decide whether they make sense, and ask useful questions to clarify or improve the arguments. Students engage in collaborative discussions, respond thoughtfully to diverse perspectives and approaches, and qualify their own views in light of evidence presented.

4. Model with mathematics.

Mathematically proficient high school students can apply the mathematics they know to solve problems arising in everyday life, society, and the workplace. By high school, a student might use geometry to solve a design problem or use a function to describe how one quantity depends on another. Mathematically proficient students who can apply what they know are comfortable making assumptions and approximations to simplify a complicated situation, realizing that these may need revision later. They are able to identify important quantities in a practical situation and map their relationships using such tools as diagrams, two-way tables, graphs, flowcharts and formulas. They can analyze those relationships mathematically to draw conclusions. They routinely interpret their mathematical results in the context of the situation and reflect on whether the results make sense, possibly improving the model if it has not served its purpose.

5. Use appropriate tools strategically.

Mathematically proficient high school students consider the available tools when solving a mathematical problem. These tools might include pencil and paper, concrete models, a ruler, a protractor, a calculator, a spreadsheet, a computer algebra system, a statistical package, or dynamic geometry software. Proficient students are sufficiently familiar with tools appropriate for high school to make sound decisions about when each of these tools might be helpful, recognizing both the insight to be gained and their limitations. For example, mathematically proficient high school students analyze graphs of functions and solutions generated using a graphing calculator. They detect possible errors by strategically using estimation and other mathematical knowledge. They are able to use technological tools to explore and deepen their understanding of concepts. When making mathematical models, they know that technology can enable them to visualize the results of varying assumptions, explore consequences, and compare predictions with data. Mathematically proficient students are able to identify relevant external mathematical resources, such as digital content located on a website, and use them to pose or solve problems.

6. Attend to precision.

Mathematically proficient high school students communicate precisely to others both verbally and in writing, adapting their communication to specific contexts, audiences, and purposes. They develop the habit of using precise language, not only as a mechanism for effective communication, but also as a tool for understanding and solving problems. Describing their ideas precisely helps students understand the ideas in new ways. They use clear definitions in discussions with others and in their own reasoning. They state the meaning of the symbols that they choose. They are careful about specifying units of measure, labeling axes, defining terms and variables, and calculating accurately and efficiently with a degree of precision appropriate for the problem context. They develop logical claims and counterclaims fairly and thoroughly in a way that anticipates the audiences' knowledge, concerns, and possible biases. High school students draw specific evidence from informational sources to support analysis, reflection, and research. They critically evaluate the claims, evidence and reasoning of others and attend to important distinctions with their own claims or inconsistencies in competing claims. Students evaluate the conjectures and claims, data, analysis, and conclusions in texts that include quantitative elements, comparing those with information found in other sources.

7. Look for and make use of structure.

Mathematically proficient high school students look closely to discern a pattern or structure. In the expression $x^2 + 9x + 14$, high school students can see the 14 as 2×7 and the 9 as $2 + 7$. They recognize the significance of an existing line in a geometric figure and can use the strategy of drawing an auxiliary line for solving problems. They also can step back for an overview and shift perspective. They can see complicated things, such as some algebraic expressions, as single objects or as being composed of several objects. For example, they can see $5 - 3(x - y)^2$ as 5 minus a positive number times a square, and use that to realize that its value cannot be more than 5 for any real numbers x and y .

8. Look for and express regularity in repeated reasoning.

Mathematically proficient high school students notice if calculations are repeated, and look both for general methods and for shortcuts. Noticing the regularity in the way terms sum to zero when expanding $(x - 1)(x + 1)$, $(x - 1)(x^2 + x + 1)$, and $(x - 1)(x^3 + x^2 + x + 1)$ might lead students to the general formula for the sum of a geometric series. As they work to solve a problem, mathematically proficient students maintain oversight of the process, while attending to the details and continually evaluating the reasonableness of their intermediate results.

WIDA English Language Development Standard 3 Language for Mathematics

English language learners communicate information, ideas and concepts necessary for academic success in the content area of mathematics.

Language Expectations: Multilingual learners will...

EXPLAIN

ELD-MA 9-12 Explain Interpretive Interpret mathematical explanations by • Identifying concept or entity • Analyzing data and owning problem-solving approaches • Evaluating rationales, models, and/or interpretations based on evidence and mathematical principles	ELD-MA 9-12 Explain Expressive Construct mathematical explanations that • Introduce mathematical concept or entity • Share solutions with others • Describe data and/or approach used to solve a problem • State reasoning used to generate own or alternate solutions
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ARGUE

ELD-MA 9-12 Argue Interpretive Interpret concepts in arguments by • Comparing conjectures with previously established results and stated assumptions • Distinguishing correct from flawed logic • Evaluating relationships among evidence and mathematical principles to create generalizations	ELD-MA 9-12 Argue Expressive Construct mathematics arguments that • Create precise conjecture, using definitions, previously established results, and stated assumptions • Generalize logical relationships across cases • Justify (and refute) conclusions with evidence and mathematical principles • Evaluate and extend others' arguments
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WIDA ELD STANDARDS FRAMEWORK 2020

HELPFUL RESOURCE FROM THE NEVADA DEPARTMENT OF EDUCATION
Integrating ELD (English Language Development) into the Standards for Mathematical Practices
(SAMPLE)

Click here for [additional resources](#) from the Nevada Department of Education

ELD STANDARDS FRAMEWORK FOR DEVELOPING THE LANGUAGE OF MATH GRADES 9-12

Section 3B: Math Disciplinary Practices

Practice 1a: Make Sense of Problems and Persevere in Solving Them – Teacher Moves

Teacher Moves: What supports can teachers provide students at different proficiency levels to use language to interpret or make meaning of the content? Examples:

Entering/Emerging (Levels 1-2)	Developing/Expanding (Levels 3-4)	Bridging/Reaching (Levels 5-6)
● Provide scaffolded tasks for students to draw a picture of their problem and of their solution and to label it. ● Provide simple patterned oral and written sentence frames for students to emulate/copy basic content provided with a predetermined learning partner. ● Model the language of mathematical expression ratio examples, and then provide the task for students to label the mathematical expressions with a predetermined learning partner; have students state the academic vocabulary associated with the number or illustrated expression. ● Provide sentence frames or sentence starters for students to use to access group discussion. For example: The first thing I did was _____. I already know _____ so _____. ● Question prompts for students: Can you please repeat that? (NEPF – IP.1.2; 2.1; 2.2; 3.1; 3.2; 5.3)	● Provide learning tasks in which students can use illustrations or numbers to explain their understanding of problems and strategies for solutions. ● Model a predetermined dialogue structure for students to state and clarify their reasoning to a partner or small group and listen to the ideas of others to agree or disagree with reasons to ensure the participation of all students. ● Provide students with sentence starters from a leveled list of scaffolding statements. For example: I think _____ because _____. I solved the problem by _____. ● Question prompts for students: Can you say it another way? So, does that mean _____? (NEPF – IP.1.2; 2.1; 2.2; 3.1; 3.2; 5.3)	● Provide learning tasks in which students can use illustrations or numbers to explain their understanding. ● Model a predetermined dialogue structure for students to state and clarify their reasoning while providing evidence to a partner or small group. Listen to the ideas of others and then agree or disagree by providing reasons to ensure the participation of all students. ● Provide students with sentence starters from a leveled list of statements using more complex language. For example: My answer makes sense because _____. I hadn't thought about _____. ● Question prompts for students: What's another way to look at the problem/ solution? What part do you understand? (NEPF – IP.1.2; 2.1; 2.2; 3.1; 3.2; 5.3)

Connecting Social Emotional Learning and Mathematics

Definition of Social Emotional Learning: Social and emotional learning (SEL) is the process through which children and adults acquire and seek to apply the knowledge, skills, and attitudes to develop healthy identities, manage emotions and achieve personal and collective goals, feel and show empathy for others, establish and maintain supportive relationships and make responsible and caring decisions. CASEL 2020

In other words, Social Emotional Learning means creating a classroom and school environment where kids feel the support of the adults around them, where the learning community is welcoming to all people (families and students alike) and where students have access to the emotional resources, tools and spaces that research has shown promote academic learning and future success.

Core Competencies of SEL:

The 5 core social emotional competencies identified by CASEL are: Self-Assessment, Self-Management, Social Awareness, Relationship Skills, and Responsible decision making. These competencies must be woven into academic learning both implicitly (through structure, learning ritual, environment and routine) and explicitly (through the physical space of the classroom, the lessons taught by educators, and the learning strategies employed by students to absorbing and producing their academic work).

SEL Practices in the classroom:

Each educator creates or enhances a culture of social-emotionally accessible learning in the following ways (Taken from Transforming Education):

- Physical space and resources available to students
- Teaching tools and student-centered discipline
- Relationship Building between educators and students
- Explicit SEL instruction within academic units
- Practice and modeling opportunities

Resources:

Mass Department of Education:

Educator Effectiveness Guidebook for Inclusive Practice: <https://www.doe.mass.edu/edeval/guidebook/>

CASEL:

Integration of SEL in Academic Instruction:

<https://schoolguide.casel.org/focus-area-3/classroom/integration-of-sel-and-instruction/>

Transforming Education:

Revised SEL integration Approach:

<https://transformingeducation.org/resources/revised-sel-integration-approach/>

Educator Self-Reflection Tool:

https://transformingeducation.org/wp-content/uploads/2020/10/Educator-Self-Reflection-Tool_vF.pdf

Connecting Social Emotional Learning and Mathematics

Social and Emotional Learning in Math: Attending to the development of students' social and emotional skills can help students, schools, and districts meet the expectations for teaching and learning that are described in the Massachusetts Curriculum Frameworks.

The content standards and mathematical practice standards in the recently adopted 2017 MA Curriculum Framework for Mathematics describe a vision of what it means to be a mathematically proficient person in this century. The progression of standards in the framework builds a strong foundation for students to apply their understandings and skills to solve real world problems. Students who meet the expectations of the mathematics content and practice standards not only develop conceptual understanding, procedural fluency, and problem solving ability but when coupled with a rich and supportive learning environment they can also develop the ability to persevere, collaborate with others and to reason, prove, justify, and communicate with precision. The 2017 Massachusetts Curriculum Framework for Mathematics includes the following Guiding Principle to support Social and Emotional Learning:

Guiding Principle 8: Social and emotional learning can increase academic achievement, improve attitudes and behaviors, and reduce emotional distress. Students should practice self-awareness, self-management, social awareness, responsible decision-making, and relationship skills, by, for example: collaborating and learning from others and showing respect for others' ideas; applying the mathematics they know to make responsible decisions to solve problems, engaging and persisting in solving challenging problems; and learning that with effort, they can continue to improve and be successful.

In the table below are examples of intersections between the Standards for Mathematical Practice and the SEL learning Competencies¹.

Mathematical Practice Standards (SMP)	Social and Emotional Learning Competencies
1. Make sense of problems and persevere in solving them.	Self-awareness: The ability to accurately recognize one's emotions and thoughts and their influence on behavior. This includes accurately assessing one's strengths and limitations and possessing a well-grounded sense of confidence and optimism. (SMP 1,4,7,8)
2. Reason abstractly and quantitatively.	Self-management: The ability to regulate one's emotions, thoughts, and behaviors effectively in different situations. This includes managing stress, controlling impulses, motivating oneself, and setting and working toward achieving personal and academic goals. (SMP 1,2,3,4,5,6,7,8)
3. Construct viable arguments and critique the reasoning of others.	Social awareness: The ability to take the perspective of and empathize with others from diverse backgrounds and cultures, to understand social and ethical norms for behavior, and to recognize family, school, and community resources and supports. (SMP 1,3,6)
4. Model with mathematics.	Relationship skills: The ability to establish and maintain healthy and rewarding relationships with diverse individuals and groups. This includes communicating clearly, listening actively, cooperating, resisting inappropriate social pressure, negotiating conflict constructively, and seeking and offering help when needed. (SMP 3)
5. Use appropriate tools strategically.	Responsible decision making: The ability to make constructive and respectful choices about personal behavior and social interactions based on consideration of ethical standards, safety concerns, social norms, the realistic evaluation of consequences of various actions, and the wellbeing of self and others.
6. Attend to precision.	
7. Look for and make use of structure.	
8. Look for and express regularity in repeated reasoning.	

